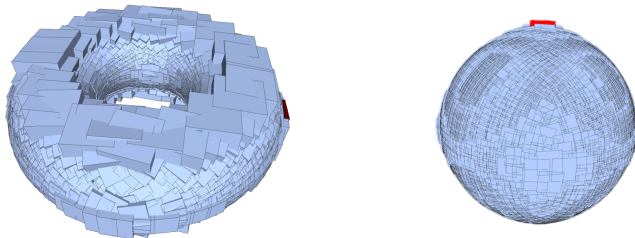


Certified Approximations of Algebraic Varieties



Krawczyk Test

Theorem

(Krawczyk point test). Let F be a system of n polynomials $F = (f_1, \dots, f_n)$ in n -variables over $\mathbb{R}$, and $x \in \mathbb{R}^n$. Then if there exists a $\rho \in (0, 1)$, Y a $n \times n$ invertible matrix and $r > 0$ such that

$$x - YF(x) + (Id - Y^{-1}JF(x + rB))rB \subseteq x + \rho rB,$$

where $B = [-1, 1]^n$ and JF denotes the Jacobian of F , then $x + rB$ contains a point on our variety $\mathcal{V}(f_1, \dots, f_n)$.